

AI-Based Crop Health Monitoring Using Drone Multispectral Data

1. Project Overview

Crop health monitoring is an essential part of precision agriculture. Early identification of crop stress helps farmers reduce yield loss, improve resource utilization, and increase productivity. In this project, drone-derived multispectral vegetation indices were used to classify crop health using machine learning techniques.

The dataset consists of processed multispectral vegetation indices including **NDVI**, **GNDVI**, **SAVI**, **EVI**, **Red Edge Index**, **Canopy Density**, and **Moisture Index**, where each record represents a spatial observation from an agricultural field. Machine learning models were trained to classify each observation as **Healthy** or **Stressed**, and the predictions were visualized using field-level stress heatmaps.

2. Objective

The primary objectives of this project are:

- To understand vegetation indices used in crop health monitoring.
- To build machine learning models for crop stress classification.
- To generate field-level stress heatmaps using spatial aggregation.
- To recommend drone inspection strategies based on stress severity.
- To compare multiple machine learning algorithms and identify the best-performing model.

3. Dataset Description

The dataset contains drone-generated multispectral vegetation indices that represent crop conditions at different field locations.

The major features used in this project include:

- **NDVI (Normalized Difference Vegetation Index)**: Measures vegetation health and vigor.
- **GNDVI (Green NDVI)**: Indicates chlorophyll content and nitrogen status.
- **SAVI (Soil Adjusted Vegetation Index)**: Reduces soil background effects while measuring vegetation health.

- **EVI (Enhanced Vegetation Index):** Provides improved vegetation monitoring in dense crop regions.
- **Red Edge Index:** Detects early crop stress before visible symptoms appear.
- **Canopy Density:** Represents crop canopy coverage and plant growth.
- **Moisture Index:** Indicates water availability and moisture stress.

The target variable classifies crops into **Healthy** and **Stressed** categories.

4. Methodology

The project followed an end-to-end machine learning pipeline:

1. Data loading and preprocessing.
2. Exploratory Data Analysis (EDA).
3. Correlation analysis and feature visualization.
4. Data scaling (for Logistic Regression and SGD).
5. Model training using:
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - Stochastic Gradient Descent (SGD)
6. Model evaluation using:
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - ROC-AUC
 - Confusion Matrix
 - 5-Fold Cross Validation
7. Spatial aggregation of predictions.
8. Generation of crop stress heatmaps.
9. Model comparison and interpretation.

5. Results

Four machine learning algorithms were evaluated to classify crop stress.

- **Logistic Regression** provided strong baseline performance with good interpretability.
- **Decision Tree** was easy to understand but showed lower generalization compared to other models.
- **Random Forest** improved prediction performance by combining multiple decision trees.
- **Stochastic Gradient Descent (SGD)** achieved the best overall performance on this dataset, with the highest accuracy, precision, recall, and F1-score.

The comparison indicates that SGD is the most suitable model for crop stress classification using the available vegetation indices.

6. Spatial Analysis

The predicted crop stress values were aggregated into spatial grid cells to generate a field-level stress heatmap.

The heatmap helps identify:

- Healthy crop regions.
- Moderately stressed areas.
- Highly stressed regions requiring immediate attention.

This visualization supports precision agriculture by enabling farmers to focus resources only on affected areas.

7. Drone & Agronomy Interpretation

The stress maps generated from the machine learning models assist in planning drone operations.

- Healthy areas require routine monitoring every 1–2 weeks.
- Moderately stressed areas should be inspected every 3–5 days.
- Highly stressed regions require immediate drone inspection and field verification.

Based on vegetation indices, farmers can identify water stress, nutrient deficiency, disease symptoms, or poor canopy development and take corrective actions such as targeted irrigation, fertilizer application, or pest management.

8. Conclusion

This project demonstrates the successful application of machine learning and drone-derived multispectral vegetation indices for crop health monitoring. The developed system accurately classifies healthy and stressed crops and generates field-level stress heatmaps that support precision agriculture.

Among the evaluated models, the **Stochastic Gradient Descent (SGD)** classifier achieved the best overall performance, making it the most effective model for this dataset. The generated visualizations enable efficient drone inspection planning and support timely agronomic interventions.

Future work can incorporate raw multispectral imagery, weather data, IoT sensors, and advanced deep learning models to further improve prediction accuracy and real-time crop monitoring capabilities.